

LionX Migration

Replatforming MNY to a cloud-native foundation on AWS.



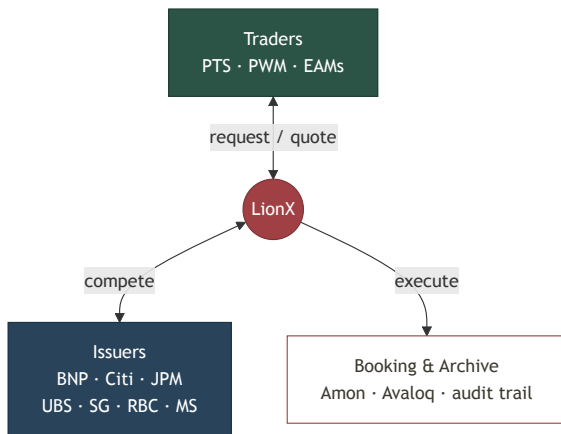
On-prem today



AWS cloud-native

What is LionX

Pictet's trading platform for structured products (internal name: MNY).



Connect

one protocol to every issuer



Quote

issuers compete, best execution



Execute

routed to internal booking (Amon, Avaloq)



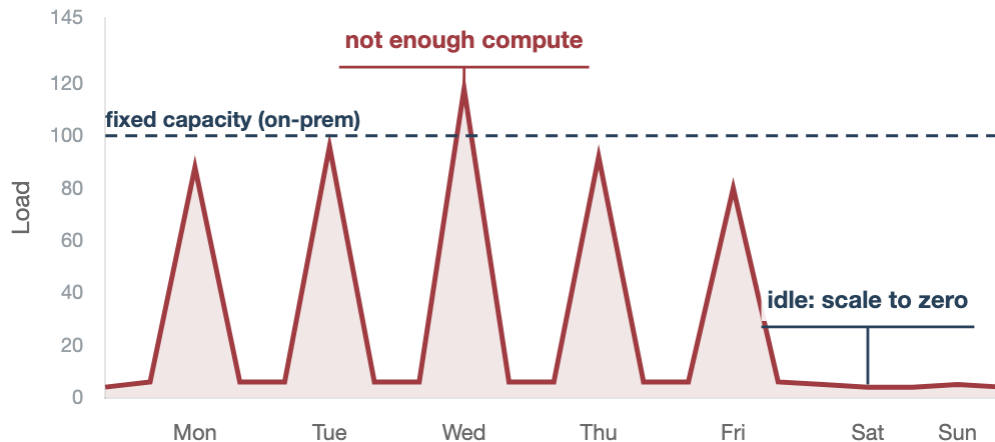
Archive

every quote and trade, full audit trail

One interface for traders. One integration point for issuers, across PTS, PWM and EAMs.

Why move to AWS

Our load peaks on market data and sits near zero off-hours and weekends.



Pictet's cloud strategy

LionX aligns with the group's move to AWS



Simplify

on-prem infra back to industry standards



Cloud-native

AWS flexibility, autoscaling on peaks



Full control

over development, operations and infrastructure

Fixed on-prem capacity sits idle most of the week, yet still falls short at peaks. AWS scales up on spikes and down to zero when idle.

Today's infrastructure limits

The on-prem stack works, but it shows its limits. AWS addresses each one.



Single-node MongoDB

reliability and architecture at risk

- ✓ **Managed, Multi-AZ, resilient and elastic**



No horizontal scaling

several components cannot absorb peaks

- ✓ **Elastic, scales with demand, no fixed capacity**



No zero-downtime deploys

PROD is stopped for every upgrade

- ✓ **Blue/green rollouts, zero downtime**



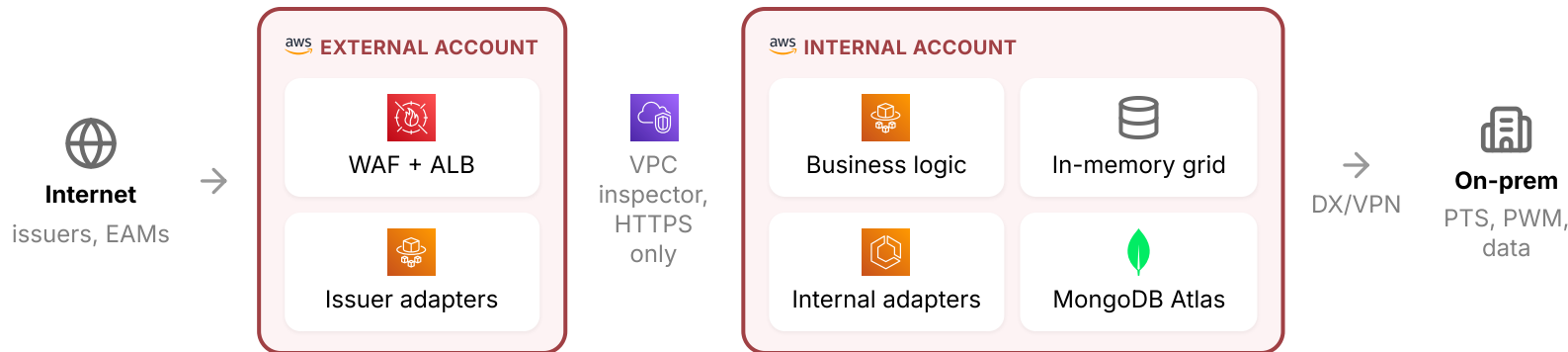
Pipeline to simplify

Puppet / Docker / Kubernetes ⇒ Terraform

- ✓ **One Terraform IaC, simple CI/CD**

Target architecture, step 1: replatform

Not a lift-and-shift: we retire the hybrid setup for managed services (Atlas, ALB) and a single deployment model.



Issuers and EAMs enter through the external account. PTS and PWM traders reach the business logic from on-prem, over the private link.

Each building block will be validated with an ADR

Shared across both accounts



CloudWatch



Secrets Manager



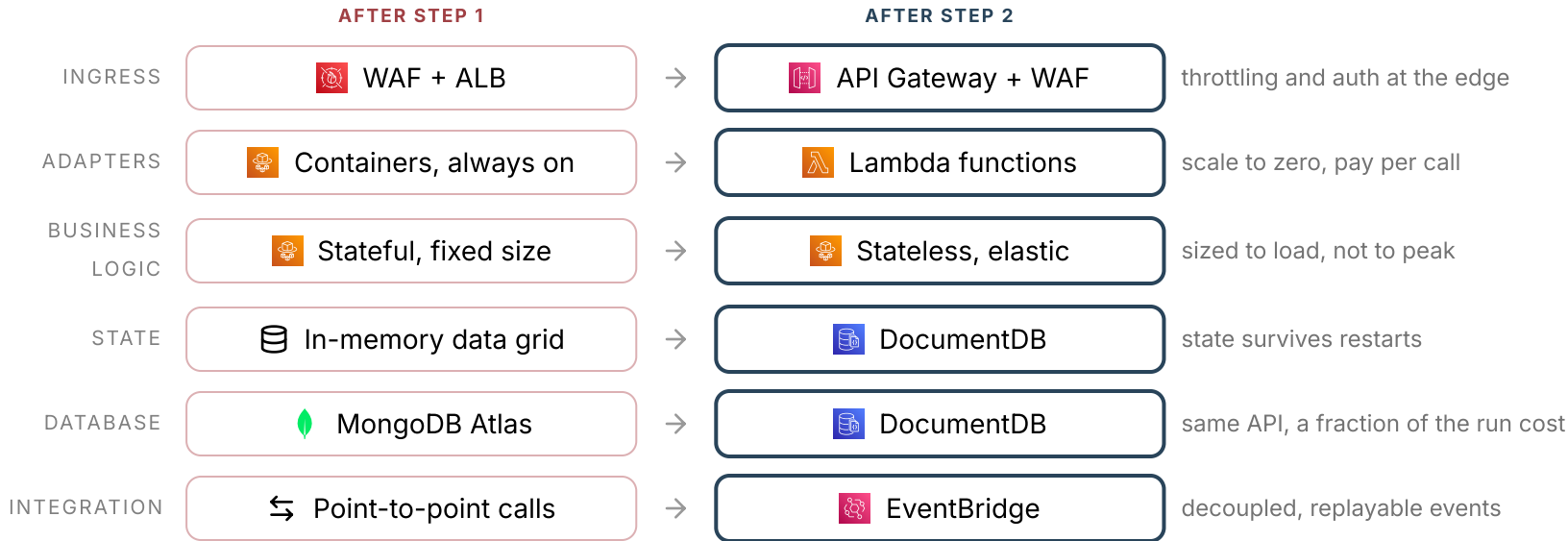
ACM PCA



S3 snapshots

Target architecture, step 2: cloud-native

Same accounts, same flows, same security model as step 1. The refactor changes six things inside them.



Everything else carries over from step 1: the two accounts, the VPC inspector, DX/VPN to on-prem, observability.

 Each change will be validated with an ADR

A two-phase migration

Business continuity first, cloud-native gains next.

PHASE 1: REPLATFORM



Iso-functional, simpler stack

same app behavior, managed services replace the hybrid setup



Parity

performance, security, resilience and audit at least equivalent



Modern operations

single push-based CI/CD, IaC, blue/green deploys

PHASE 2: LEVERAGE THE CLOUD



Elasticity

autoscaling on demand, scale down to 20% on nights and weekends



High availability

multi-AZ, self-healing, automated DR and failover tests



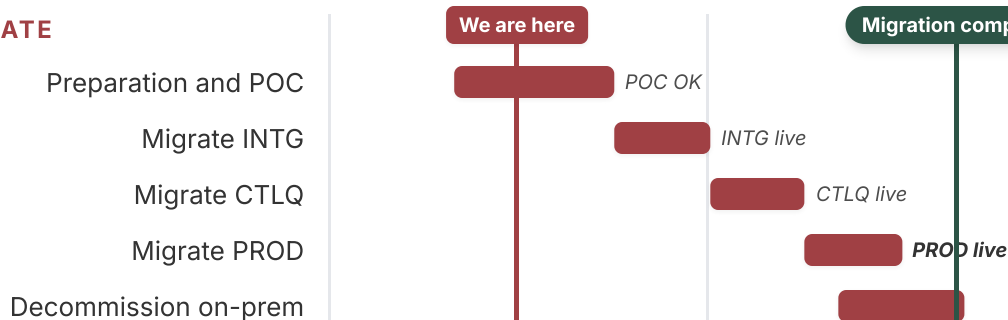
Serverless and cloud-native

FaaS adapters on Lambda, DocumentDB, no in-memory database

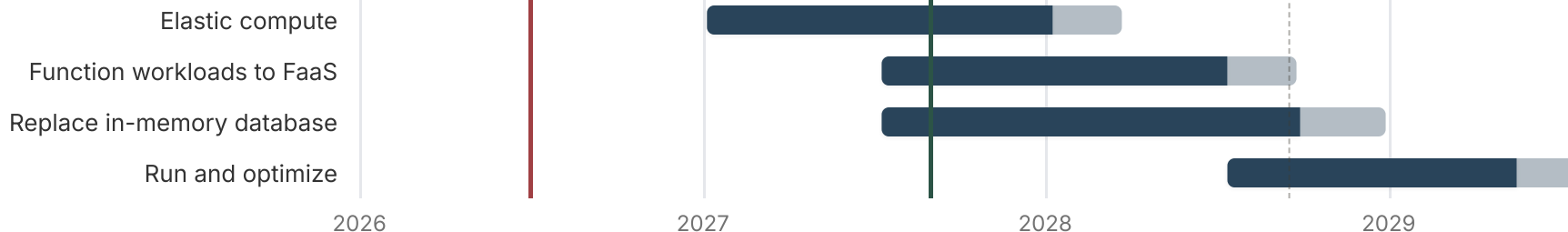
Roadmap

An estimated timeline: migrate first to reach parity, then modernize. Phased from 2026 to mid 2029.

MIGRATE



MAKE LIONX CLOUD NATIVE



Phase 1: migrate Phase 2: cloud-native

First environment live by end 2026, production on AWS by mid 2027.



Risks we are managing

Known upfront, with mitigation.



Shared Pictet services

their maturity paces us, possible temporary dual monitoring



Network complexity

on-prem links, issuer connectivity, firewall inspection



Replatforming surprises

different IO, network and timeout behavior in the cloud



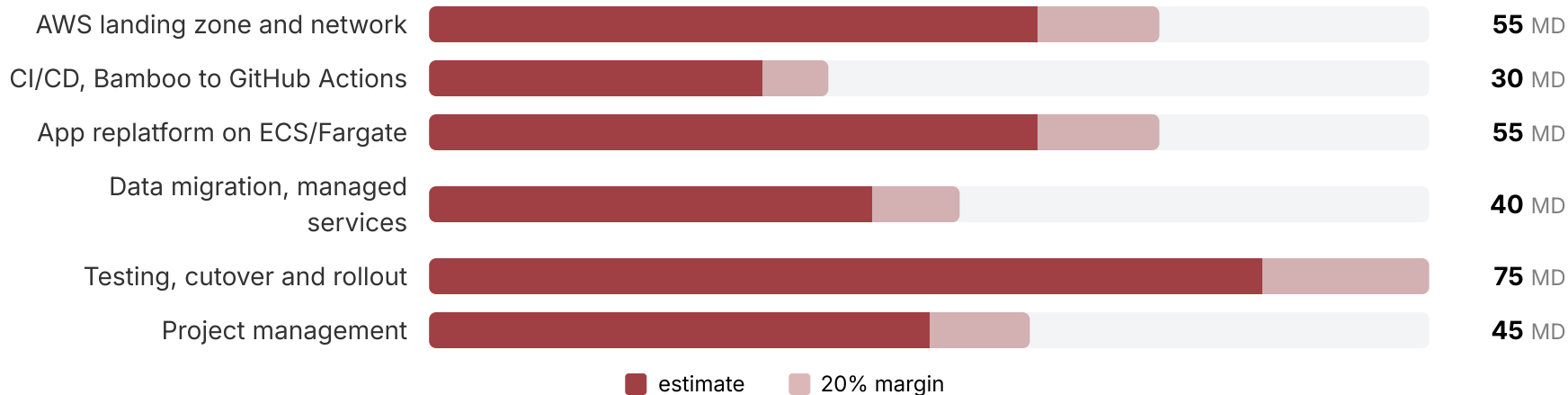
Service continuity

adapting operations and on-call during ramp-up

Each risk has an owner and a mitigation tracked in the migration plan.

Migration effort: Phase 1

Estimated one-time effort to replatform onto AWS, CI/CD from Bamboo to GitHub included.

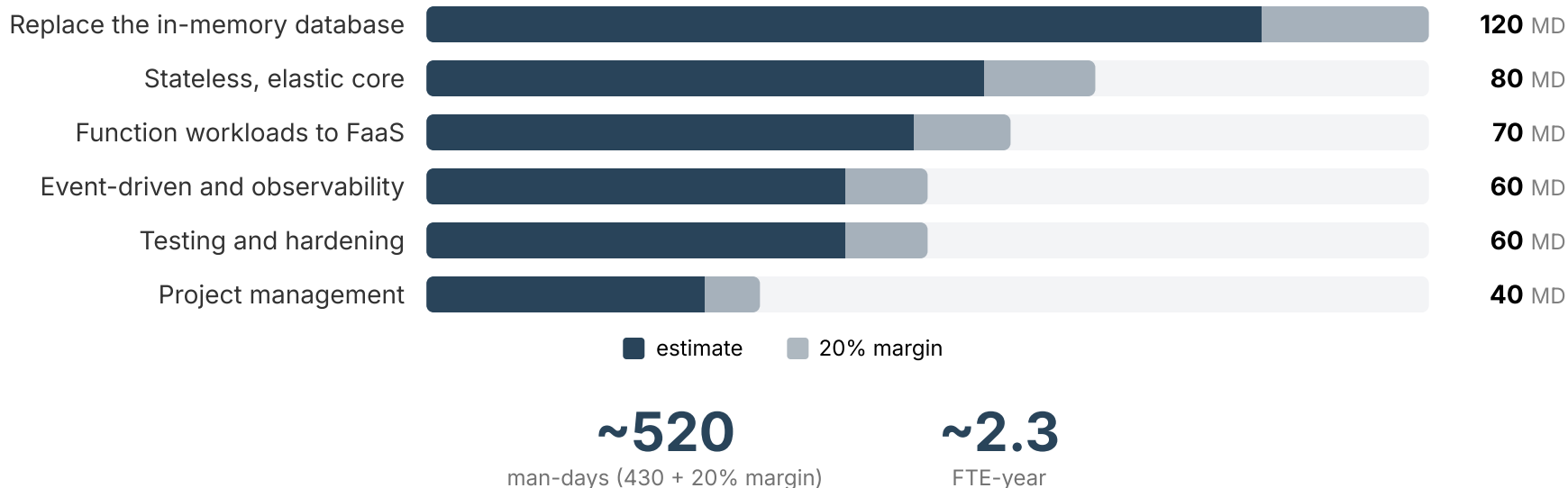


~360
man-days (300 + 20% margin)

~1.6
FTE-year

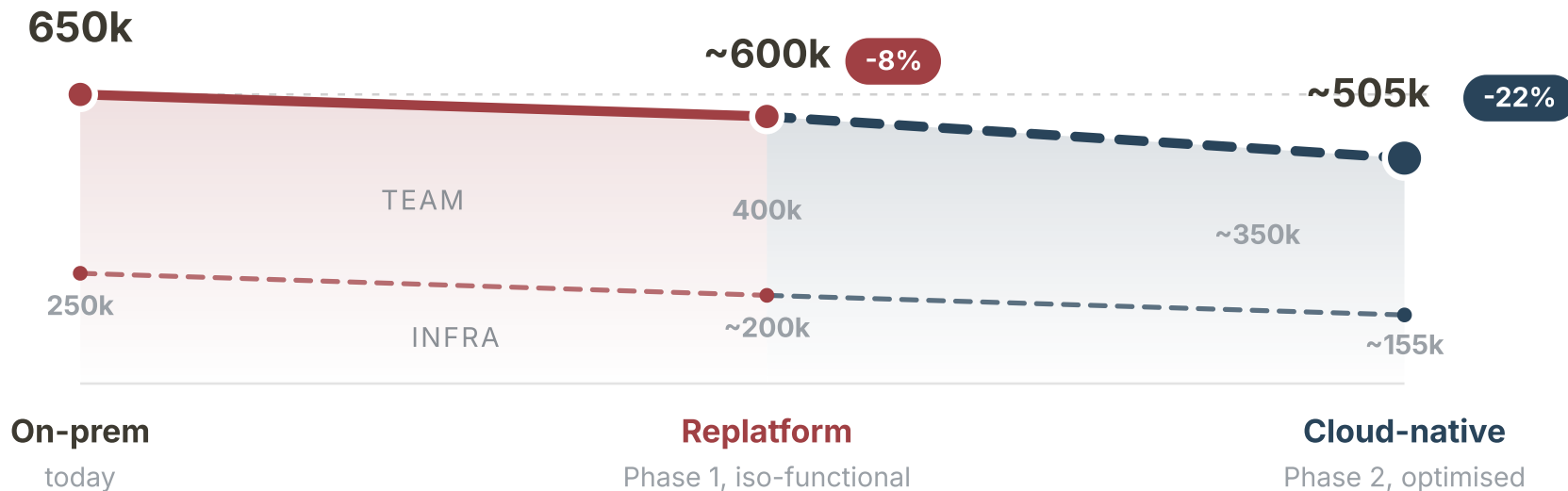
Migration effort: Phase 2

Development effort for the cloud-native refactor. A rougher estimate than Phase 1, higher uncertainty.



The cost curve

An estimate of the run cost, in CHF per year, from Zurich list prices, plus 25% contingency.



Next steps

What we are asking for today.



Green-light the plan

approve the two-phase migration
and its budget envelope



Chain the ADRs

record the architecture decisions
back to back to validate and start
fast



Integration env by end 2026

running on AWS inside a real
deployment pipeline

Questions and discussion welcome